

Chapter 4 Test model

- **70 Multiple Choice Questions (MCQs)**
 - **30 True/False Questions**
 - **Difficulty: 70% Hard, 30% Very Hard**
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Multiple Choice Questions (1–70)

1. In EER modeling, a subclass that inherits from more than one superclass is known as a:
 - a. Disjoint subclass
 - b. Shared subclass
 - c. Predicate-defined subclass
 - d. Union type

2. The relationship "SECRETARY IS-A EMPLOYEE" exemplifies which of the following?
 - a. Generalization
 - b. Aggregation
 - c. IS-A relationship
 - d. Categorization

3. Which of the following is NOT an EER extension beyond basic ER modeling?
 - a. Attribute inheritance
 - b. Weak entities
 - c. Specialization
 - d. Categories (UNION types)

4. In an attribute-defined specialization, the attribute used to determine subclass membership is called the:
 - a. Local attribute
 - b. Inherited attribute
 - c. Defining attribute
 - d. Composite attribute

5. A specialization where an entity can belong to more than one subclass is termed:
- Total
 - Disjoint
 - Overlapping
 - Partial
-

6. In EER diagrams, a double line connecting a superclass to its subclasses indicates:
- Disjoint constraint
 - Overlapping constraint
 - Total completeness
 - Partial completeness
-

7. Which symbol is used in EER diagrams to denote a disjoint constraint?
- $\circ$
 - d
 - $\cap$
 - $\cup$
-

8. The process of defining subclasses of a superclass based on distinguishing characteristics is called:
- Generalization
 - Abstraction
 - Specialization
 - Categorization
-

9. A category (UNION type) differs from a shared subclass in that a category member must exist in:
- All of its superclasses
 - None of its superclasses
 - At least one of its superclasses
 - Exactly two of its superclasses
-

10. Inheritance in EER modeling allows a subclass to inherit:
- Only attributes from the superclass
 - Only relationships from the superclass

- c. Both attributes and relationships from the superclass
 - d. Neither attributes nor relationships
-

11. Which of the following best describes generalization?

- a. Top-down refinement
 - b. Bottom-up synthesis
 - c. Horizontal partitioning
 - d. Vertical fragmentation
-

12. If a specialization uses the same defining attribute for all its subclasses, it is called:

- a. User-defined
 - b. Predicate-defined
 - c. Attribute-defined
 - d. Condition-independent
-

13. In a specialization lattice, a subclass may have:

- a. Only one direct superclass
 - b. Multiple direct superclasses
 - c. No superclasses
 - d. Only leaf superclasses
-

14. The term “multiple inheritance” in EER modeling refers to:

- a. A superclass inheriting from multiple subclasses
 - b. A subclass inheriting from multiple superclasses
 - c. Overlapping specializations with shared attributes
 - d. Recursive generalization
-

15. Which of the following is a valid reason for using a category instead of a shared subclass?

- a. When the superclasses represent disjoint entity types
 - b. When the subclass must be a member of all superclasses
 - c. When the subclass represents a union of heterogeneous superclasses
 - d. When inheritance must be single-parent only
-

16. In the EER model, local attributes of a subclass are:

- a. Inherited from the superclass
 - b. Shared with other subclasses
 - c. Specific to that subclass
 - d. Always composite
-

17. Consider an EMPLOYEE superclass with subclasses ENGINEER, MANAGER, and SALARIED_EMPLOYEE. An employee who is both a manager and an engineer belongs to:

- a. Only MANAGER
 - b. Only ENGINEER
 - c. Both MANAGER and ENGINEER
 - d. None, due to disjoint constraint
-

18. Which constraint ensures that every entity in the superclass is a member of at least one subclass?

- a. Disjointness
 - b. Overlapping
 - c. Total completeness
 - d. Partial inheritance
-

19. In EER diagrams, user-defined subclasses are typically indicated by:

- a. Absence of a predicate label
 - b. A "u" symbol
 - c. Dashed lines
 - d. No standard notation—implied by context
-

20. A specialization hierarchy enforces:

- a. Multiple inheritance
 - b. Single inheritance
 - c. Total completeness only
 - d. Overlapping membership
-

21. Which of the following is true about shared subclasses?

- a. They represent the union of their superclasses

- b. They must be members of all their superclasses
 - c. They cannot have local attributes
 - d. They violate the IS-A principle
-

22. The EER model supports which type of inheritance?

- a. Only single inheritance
 - b. Only multiple inheritance
 - c. Both single and multiple inheritance
 - d. Neither—inheritance is not supported
-

23. In a UNIVERSITY EER schema, if GRADUATE_STUDENT is a subclass of both STUDENT and EMPLOYEE, this is an example of:

- a. Attribute-defined specialization
 - b. Disjoint hierarchy
 - c. Shared subclass (multiple inheritance)
 - d. Category (UNION type)
-

24. Which of the following is NOT a type of specialization constraint?

- a. Disjoint, total
 - b. Overlapping, partial
 - c. Predicate, recursive
 - d. Disjoint, partial
-

25. What does the "o" symbol denote in an EER diagram?

- a. Total completeness
 - b. Overlapping subclasses
 - c. Optional participation
 - d. Object-oriented class
-

26. Generalization is typically:

- a. Partial and disjoint
- b. Total and overlapping
- c. Total because the superclass is derived from subclasses
- d. Always user-defined

-
27. In EER modeling, a subclass that does not include all superclass entities demonstrates:
- a. Total completeness
 - b. Partial completeness
 - c. Disjointness
 - d. Attribute inheritance
-

28. Which of the following best describes a predicate-defined subclass?
- a. Membership is assigned by the DBA per entity
 - b. Membership is determined by a condition on a superclass attribute
 - c. Membership requires intersection of multiple superclasses
 - d. Membership is based on external ontologies
-

29. In the vehicle registration example, OWNER is a category because:
- a. It inherits from VEHICLE
 - b. It is a shared subclass of PERSON, BANK, and COMPANY
 - c. It represents the union of PERSON, BANK, and COMPANY
 - d. It is a superclass of REGISTERED_VEHICLE
-

30. Local relationships in a subclass are:
- a. Inherited from the superclass
 - b. Available to all subclasses
 - c. Specific to that subclass only
 - d. Always total participation
-

31. Which statement is correct regarding inheritance in EER?
- a. Subclasses inherit only key attributes
 - b. Superclasses inherit local attributes from subclasses
 - c. Subclasses inherit all attributes and relationships of the superclass
 - d. Inheritance applies only to relationships, not attributes
-

32. A specialization with defining attribute JobType and subclasses {SECRETARY, ENGINEER, TECHNICIAN} is:
- a. User-defined and disjoint

- b. Attribute-defined and possibly overlapping
 - c. Predicate-defined and total
 - d. Category-based
-

33. In a lattice structure, a subclass can:
- a. Have only one ancestor
 - b. Participate in multiple IS-A hierarchies
 - c. Not inherit from more than one level
 - d. Only be leaf-level
-

34. The primary purpose of a category (UNION type) is to:
- a. Enforce disjointness
 - b. Model a subclass of heterogeneous superclasses
 - c. Replace generalization
 - d. Simplify attribute inheritance
-

35. Which of the following is a key difference between generalization and specialization?
- a. Generalization is top-down; specialization is bottom-up
 - b. Specialization creates superclasses; generalization creates subclasses
 - c. Generalization is bottom-up; specialization is top-down
 - d. They are identical in EER modeling
-

36. In EER diagrams, how is an attribute-defined specialization typically displayed?
- a. With the defining attribute name near the superclass-subclass line
 - b. With a double diamond
 - c. With a "U" symbol
 - d. With dashed ellipses
-

37. If an EMPLOYEE can be both HOURLY_EMPLOYEE and SALARIED_EMPLOYEE, the specialization must be:
- a. Disjoint and total
 - b. Overlapping and partial
 - c. Disjoint and partial
 - d. Overlapping and total

-
38. Which of the following is true about the IS-A relationship?
- a. It implies the subclass is a different real-world entity
 - b. It allows an entity to exist only in the subclass
 - c. The subclass and superclass represent the same real-world entity in different roles
 - d. It prohibits attribute inheritance
-

39. In a shared subclass like ENGINEERING_MANAGER (subclass of both MANAGER and ENGINEER), the entity must:
- a. Belong to either MANAGER or ENGINEER
 - b. Belong to both MANAGER and ENGINEER
 - c. Belong to neither
 - d. Be a category
-

40. What is the inheritance path in a specialization lattice?
- a. Only from immediate superclass
 - b. From all predecessor superclasses
 - c. Only from the root superclass
 - d. No inheritance in lattices
-

41. Which of the following constraints is most likely applied to a generalization?
- a. Partial and disjoint
 - b. Overlapping and partial
 - c. Total and disjoint
 - d. Total (due to derivation from subclasses)
-

42. A subclass with no local attributes or relationships is still meaningful if:
- a. It enables constraint modeling (e.g., disjointness)
 - b. It has a composite key
 - c. It participates in a category
 - d. It is user-defined
-

43. In EER modeling, the term "class" can refer to:
- a. Only superclasses

- b. Only subclasses
 - c. Entity types, superclasses, or subclasses
 - d. Only categories
-

44. Which of the following is an example of a user-defined subclass?
- a. EMPLOYEE with JobType = 'Engineer'
 - b. STUDENT under Age > 25
 - c. A DBA manually assigning an employee to a "VIP" subclass
 - d. CAR and TRUCK generalized to VEHICLE
-

45. The completeness constraint in specialization controls:
- a. Whether subclasses overlap
 - b. Whether superclass entities must belong to a subclass
 - c. Whether inheritance is single or multiple
 - d. Whether attributes are atomic
-

46. In an EER diagram, a single line from superclass to subclasses indicates:
- a. Total completeness
 - b. Partial completeness
 - c. Disjointness
 - d. Overlapping
-

47. Which of the following scenarios requires a category rather than a shared subclass?
- a. A GRADUATE_STUDENT who is both STUDENT and EMPLOYEE
 - b. An OWNER who can be a PERSON, BANK, or COMPANY
 - c. A TECHNICIAN who is also SALARIED_EMPLOYEE
 - d. A CAR that is a VEHICLE
-

48. Attribute inheritance implies that:
- a. Subclass attributes override superclass attributes
 - b. Every subclass entity has values for all inherited attributes
 - c. Superclasses gain attributes from subclasses
 - d. Only key attributes are inherited
-

49. In a disjoint, total specialization:

- a. Every superclass entity belongs to exactly one subclass
 - b. Entities may belong to multiple subclasses
 - c. Some entities may belong to no subclass
 - d. Membership is user-assigned
-

50. The EER model was developed primarily to:

- a. Replace relational modeling
 - b. Support object-oriented databases
 - c. Enhance ER modeling with subclassing and inheritance
 - d. Simplify SQL queries
-

51. Which of the following is a characteristic of a predicate-defined subclass?

- a. Defined by DB user per instance
 - b. Defined by a condition like Age > 30
 - c. Requires multiple inheritance
 - d. Cannot coexist with attribute-defined subclasses
-

52. In the absence of any constraint notation in an EER diagram, the default assumption is:

- a. Disjoint and total
 - b. Overlapping and partial
 - c. Disjoint and partial
 - d. Overlapping and total
-

53. A specialization hierarchy differs from a lattice in that a hierarchy:

- a. Allows multiple inheritance
 - b. Is always total
 - c. Enforces single inheritance (tree structure)
 - d. Cannot have shared subclasses
-

54. The subclass MANAGER in an EMPLOYEE specialization based on role is likely:

- a. Attribute-defined with defining attribute Role
- b. User-defined
- c. Predicate-defined with condition Role = 'Manager'
- d. Any of the above, depending on design choice

55. Which of the following is NOT inherited by a subclass?

- a. Primary key of superclass
 - b. Composite attributes of superclass
 - c. Local attributes of another subclass
 - d. Relationships of superclass
-

56. In EER modeling, a “shared subclass” leads to:

- a. Union types
 - b. Multiple inheritance
 - c. Disjoint constraints
 - d. Partial completeness
-

57. Categories are particularly useful when superclasses are:

- a. Homogeneous
 - b. Subclasses of the same root
 - c. Heterogeneous (e.g., PERSON, COMPANY)
 - d. Predicate-defined
-

58. Which EER concept allows modeling an entity that plays multiple distinct roles?

- a. Weak entity
 - b. Aggregation
 - c. Subclassing with overlapping specialization
 - d. Derived attribute
-

59. In a total specialization, if an entity is inserted into the superclass, the DBMS must:

- a. Reject the insertion
 - b. Automatically assign it to a subclass
 - c. Allow it only if it fits a subclass condition
 - d. Require explicit subclass assignment (depending on implementation)
-

60. The notation for a category in EER diagrams typically includes:

- a. A double rectangle
- b. A circle labeled “U” or “UNION”

- c. A dashed oval
 - d. No standard symbol—contextual
-

61. Which of the following best describes the semantic difference between intersection (shared subclass) and union (category)?
- a. Shared subclass = OR; Category = AND
 - b. Shared subclass = AND; Category = OR
 - c. Both represent AND
 - d. Both represent OR
-

62. In an overlapping, partial specialization of EMPLOYEE into {ENGINEER, MANAGER}, an employee who is neither is:
- a. Invalid
 - b. Allowed
 - c. Automatically assigned to a default subclass
 - d. Converted to a weak entity
-

63. Generalization is often considered a ___ process.
- a. Top-down
 - b. Deductive
 - c. Bottom-up
 - d. Recursive
-

64. Specialization is often considered a ___ process.
- a. Inductive
 - b. Bottom-up
 - c. Top-down
 - d. Aggregative
-

65. Which of the following is a very hard distinction in EER modeling?
- a. Between entity and attribute
 - b. Between specialization and generalization in diagrams
 - c. Between strong and weak entity
 - d. Between simple and composite attribute

-
66. In a lattice, if class C inherits from A and B, and A and B both inherit from ROOT, then C inherits:
- a. Only from A and B
 - b. From A, B, and ROOT
 - c. Only from ROOT
 - d. Nothing due to ambiguity
-

67. The EER model handles which of the following that basic ER cannot?
- a. Binary relationships
 - b. Cardinality constraints
 - c. Subclass-specific attributes and multiple inheritance
 - d. Weak entities
-

68. If a specialization is both disjoint and total, the number of subclasses an entity can belong to is:
- a. Zero
 - b. One
 - c. One or more
 - d. All
-

69. Which of the following is a potential issue with multiple inheritance in EER?
- a. Attribute name conflict from different superclasses
 - b. Violation of first normal form
 - c. Inability to represent relationships
 - d. Loss of key constraints
-

70. In the absence of explicit constraints, EER modeling assumes subclass membership is:
- a. Disjoint and total
 - b. Overlapping and total
 - c. Overlapping and partial
 - d. Disjoint and partial
-

True/False Questions (71–100)

71. In EER modeling, an entity can exist solely as a member of a subclass without being in the superclass.

- A. True
 - B. False
-

72. Attribute inheritance means a subclass automatically gets all attributes of its superclass.

- A. True
 - B. False
-

73. A category is a subclass that is a subset of the intersection of its superclasses.

- A. True
 - B. False
-

74. Generalization is the reverse process of specialization.

- A. True
 - B. False
-

75. In a disjoint specialization, an entity may belong to more than one subclass.

- A. True
 - B. False
-

76. The "d" symbol in EER diagrams denotes overlapping subclasses.

- A. True
 - B. False
-

77. A shared subclass must be a member of all its superclasses.

- A. True
 - B. False
-

78. User-defined subclasses use conditions based on attribute values.

- A. True

B. False

79. In a total specialization, every entity in the superclass must belong to at least one subclass.

A. True

B. False

80. Specialization hierarchies allow multiple inheritance.

A. True

B. False

81. A lattice structure in EER modeling supports single inheritance only.

A. True

B. False

82. Local attributes of a subclass are not inherited by its own subclasses.

A. True

B. False

83. The IS-A relationship implies identity of the real-world entity across superclass and subclass.

A. True

B. False

84. Categories are used when superclasses are semantically similar.

A. True

B. False

85. In EER diagrams, a double line indicates partial completeness.

A. True

B. False

86. Predicate-defined subclasses are a form of condition-defined specialization.

- A. True
 - B. False
-

87. Generalization typically results in a total constraint.

- A. True
 - B. False
-

88. A subclass can participate in relationships that its superclass does not.

- A. True
 - B. False
-

89. All subclasses in an attribute-defined specialization must use the same defining attribute.

- A. True
 - B. False
-

90. Inheritance in EER includes both attributes and relationships from all ancestor superclasses.

- A. True
 - B. False
-

91. The EER model eliminates the need for foreign keys.

- A. True
 - B. False
-

92. An overlapping specialization can also be total.

- A. True
 - B. False
-

93. A category member must exist in exactly one of its superclasses.

- A. True
 - B. False
-

94. Specialization is a bottom-up design process.

- A. True
 - B. False
-

95. In a UNIVERSITY EER schema, RESEARCH_ASSISTANT as a subclass of both STUDENT and EMPLOYEE is a shared subclass.

- A. True
 - B. False
-

96. The defining attribute in an attribute-defined specialization must be single-valued.

- A. True
 - B. False
-

97. EER modeling does not support recursive subclassing.

- A. True
 - B. False
-

98. In a disjoint, partial specialization, an entity may belong to zero or one subclass.

- A. True
 - B. False
-

99. Categories are represented using the same notation as shared subclasses in EER diagrams.

- A. True
 - B. False
-

100. The EER model is an extension of the basic ER model.

- A. True
 - B. False
-

Answer Key

Multiple Choice (1–70)

- 1. b
- 2. c
- 3. b
- 4. c
- 5. c
- 6. c
- 7. b
- 8. c
- 9. c
- 10. c
- 11. b
- 12. c
- 13. b
- 14. b
- 15. c
- 16. c
- 17. c
- 18. c
- 19. d
- 20. b
- 21. b
- 22. c
- 23. c
- 24. c
- 25. b
- 26. c
- 27. b
- 28. b
- 29. c
- 30. c
- 31. c

32. b

33. b

34. b

35. c

36. a

37. b

38. c

39. b

40. b

41. d

42. a

43. c

44. c

45. b

46. b

47. b

48. b

49. a

50. c

51. b

52. b

53. c

54. d

55. c

56. b

57. c

58. c

59. d

60. b

61. b

62. b

63. c

64. c

65. b

66. b

67. c

68. b

69. a

70. c

True/False (71–100)

- 71. B
 - 72. A
 - 73. B
 - 74. A
 - 75. B
 - 76. B
 - 77. A
 - 78. B
 - 79. A
 - 80. B
 - 81. B
 - 82. B
 - 83. A
 - 84. B
 - 85. B
 - 86. A
 - 87. A
 - 88. A
 - 89. A
 - 90. A
 - 91. B
 - 92. A
 - 93. B
 - 94. B
 - 95. A
 - 96. A
 - 97. B
 - 98. A
 - 99. B
 - 100. A
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